

Day 19 Student Handout

Geometry Summer Credit | Arc Length, Sector Area, and Radian Readiness

Name:	Date:	Period:
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Learning Targets

I can calculate arc length and sector area as fractions of a circle. I can solve for missing angle or radius. I can connect common degree measures to radians as preparation for Trigonometry.

Lesson 19A: Fraction of a Circle

Warm-Up

1.	Find the circumference of a circle with radius 8. Leave the answer in terms of pi.
2.	Find the area of a circle with radius 8. Leave the answer in terms of pi.
3.	What fraction of a circle is a 90-degree central angle?

One Fraction, Two Measurements

A central angle θ represents $\frac{\theta}{360}$ of a circle. Arc length = $(\frac{\theta}{360})(2\pi r)$. Sector area = $(\frac{\theta}{360})(\pi r^2)$. Arc length uses circumference; sector area uses circle area.

Problem	Work / notes
A circle has radius 12 and central angle 60 degrees. Find the arc length.	Fraction: Circumference: Arc length:
A circle has radius 9 and central angle 120 degrees. Find the sector area.	Fraction: Circle area: Sector area:
A circle has radius 15 and arc length 5 pi. Find the central angle.	Full circumference: Fraction of circle: Angle:

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Lesson 19B | Radian Readiness

Degrees and Radians

A New Angle Unit

One full turn is 360 degrees or 2π radians. One half-turn is 180 degrees or π radians. Convert degrees to radians by multiplying by $\pi/180$. Convert radians to degrees by multiplying by $180/\pi$.

Problem	Work / notes
Convert 45 degrees to radians.	Multiply by $\pi/180$: Simplify:
Convert $2\pi/3$ radians to degrees.	Multiply by $180/\pi$: Simplify:
Explain why 90 degrees equals $\pi/2$ radians using a fraction of a full turn.	Degree fraction: Radian fraction:

Common Reference Angles

Useful Matches

30 degrees = $\pi/6$, 45 degrees = $\pi/4$, 60 degrees = $\pi/3$, 90 degrees = $\pi/2$, 180 degrees = π , 270 degrees = $3\pi/2$, and 360 degrees = 2π .

4.	A circle has radius 10 and central angle 72 degrees. Find both arc length and sector area.
5.	Describe the key difference between arc length and sector area, including units.